

科學 HW6

$$u = t^z, du = z t^{z-1} dt$$

$$11. \times 7. \quad \Gamma(z+1) = \int_0^\infty e^{-t} t^{z+1-1} dt = \int_0^\infty e^{-t} t^z dt. \quad dv = e^{-t} dt, v = -e^{-t}$$

$$\text{By Integration by part, } \Gamma(z+1) = -t^z e^{-t} \Big|_0^\infty + z \int_0^\infty e^{-t} t^{z-1} dt = 0 + z \int_0^\infty e^{-t} t^{z-1} dt = z \Gamma(z) \quad \#$$

$$(\because \lim_{t \rightarrow \infty} -\frac{t^z}{e^t} \stackrel{\text{L'H}}{=} \lim_{t \rightarrow \infty} -\frac{z!}{e^t} = 0) \quad \swarrow$$

11. \times 9.

$$u_{n+1} = u_n + h \left[\left(1 - \frac{\alpha}{2}\right) f(x_n, u_n) + \frac{\alpha}{2} f(x_{n+1}, u_{n+1}) \right] \rightarrow (*)$$

$$\textcircled{1} \alpha=0 \Rightarrow u_{n+1} = u_n + h f(x_n, u_n) \Rightarrow \text{Forward Euler}$$

$$\textcircled{2} \alpha=1 \Rightarrow u_{n+1} = u_n + \frac{h}{2} (f_n + f_{n+1}) \Rightarrow \text{Trapezoidal method.}$$

$$\textcircled{3} \alpha=2 \Rightarrow u_{n+1} = u_n + h f(x_{n+1}, u_{n+1}) \Rightarrow \text{Backward Euler.}$$

$$\begin{aligned} \text{Consider } \gamma(x_{n+1}) &= \gamma(x_n) + h \left[\left(1 - \frac{\alpha}{2}\right) \gamma'(x_n) + \frac{\alpha}{2} \gamma'(x_{n+1}) \right] \\ &= \gamma(x_n) + h \left[\left(1 - \frac{\alpha}{2}\right) \gamma'(x_n) + \frac{\alpha}{2} (\gamma'(x_n) + h \gamma''(x_n)) \right] + O(h^3) \end{aligned}$$

$$\therefore \gamma(x_{n+1}) = \gamma(x_n) + h \gamma'(x_n) + \alpha \frac{h^2}{2} \gamma''(x_n) + O(h^3), \text{ then comparing the Taylor expansion, which}$$

$$\text{is } \gamma(x_{n+1}) = \gamma(x_n) + h \gamma'(x_n) + \frac{h^2}{2} \gamma''(x_n) + O(h^3) \Rightarrow \text{hence for } \alpha=1, \text{ it's 2nd-order consistent;}$$

and for $\alpha \neq 1$, we only have 1st-order consistent.

$$\text{For } \alpha=1, \begin{cases} \gamma'(x) = -10\gamma(x), x > 0 \\ \gamma(0) = 1 \end{cases}, \quad \begin{aligned} u_{n+1} &= u_n + \frac{h}{2} (f_n + f_{n+1}) \\ &= u_n + \frac{h}{2} (f(x_n, u_n) + f(x_{n+1}, u_{n+1})). \end{aligned}$$

$$\therefore \text{When } \gamma'(x) = -10\gamma(x) \Rightarrow u_{n+1} = u_n + \frac{h}{2} (-10u_n - 10u_{n+1})$$

$$= f(x, \gamma) \Rightarrow (1+5h)u_{n+1} = (1-5h)u_n \Rightarrow u_{n+1} = \frac{1-5h}{1+5h} u_n$$

Hence, for the I.V.P, $\lambda = -10$, $\therefore \text{Re}(-10) < 0$, then consider the trapezoid method.

$$\begin{aligned} \text{From the scheme } u_{n+1} &= u_n + h \left[\left(1 - \frac{\alpha}{2}\right) \lambda u_n + \frac{\alpha}{2} \lambda u_{n+1} \right] \\ \Rightarrow u_{n+1} \left[1 - \frac{\alpha}{2} h \lambda \right] &= u_n \left[1 + \left(1 - \frac{\alpha}{2}\right) h \lambda \right] \Rightarrow \frac{u_{n+1}}{u_n} = \frac{(1 + (1 - \frac{\alpha}{2}) h \lambda)}{1 - \frac{\alpha}{2} h \lambda} \end{aligned}$$

$$\text{take } \alpha=1, \frac{u_{n+1}}{u_n} = \frac{1-5h}{1+5h} \Rightarrow u_n = \left(\frac{1-5h}{1+5h} \right)^n, \text{ We want } \left| \frac{1-5h}{1+5h} \right| < 1.$$

• For $h > 0$, $|1-5h| < 1+5h$ is holding, so for $h > 0$, this method for the I.V.P is absolutely stable #